

Digital FIR Filter Implementation using VIVADO®

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This project implements a Finite Impulse Response (FIR) filter on the Zynq-7000 FPGA using Xilinx Vivado. FIR filters are widely used in digital signal processing due to their inherent stability, linear phase, and ease of hardware implementation. The objective is to design, simulate, and synthesize a 32-tap FIR Low-Pass Filter and verify its operation using behavioural simulation.

FIR Filter Theory:

Finite Impulse Response Filters are a form of convolution operation-based filters given by this formula:

$$y[n] = \sum_{k=0}^{N-1} (b_k \times x[n - k])$$

Where:

- b_k : filter coefficients,
- $x[n]$: input samples,
- N : number of taps.

For our project, we have implemented a 32-tap filter with the LPF.v module having the calculated coefficients for the 32 taps.

Design Architecture

- Filter Module
 - 32-tap Coefficient-Based Delay Line (Symmetric Coefficients for LPF and Antisymmetric Coefficients for the HPF)
 - A MAC operation computing the filter output
 - A Signal Delay parameter altering the address and Data Lines.
- ROM Module
 - 401 sample Dataset of a sinusoidal wave whose frequency can be altered using the address update line
 - Python Script to generate the ROM Module using CSV data of the Waveform.
- Final Filter Top Model

- A top model that combines all the data and outputs the filtered data.
 - Filter operation being Low-Pass and High-Pass from the respective modules
 - Band-Pass and Band-Stop using series and parallel combination of module output data.
- VIO and ILA
 - Using Designated VIO (Virtual Input Output) and ILA (Integrated Logic Analyzer) modules to output the waveform data using the FPGA
 - The modules are fed the Data along with specifications of the Depth of output to give us an accurate filtered output.

Timing, Constraints and Usage

- LUT Usage:

Name	Slice LUTs	Slice Regs	F7 Mux	F8 Mux	Logic LUTs	Memory LUTs	Block RAMs	DSP48E1s
Filter Modules	2038	1805	0	0	2038	0	0	48
Final Filter	3855	4637	50	16	3684	171	37.5	48
VIO/ILA	1337	2289	50	16	1190	147	37	0

- Timing Summary:
 - Setup:
 - Worst Negative Slack = 29.559 ns
 - Total Number of Endpoints = 1041
 - Hold:
 - Worst Hold Slack = 0.0093 ns
 - Total Number of Endpoints = 1033
 - Pulse Width:
 - Worst Pulse Width Slack = 15.250 ns
 - Total Number of Endpoints = 487
 - Period = 33 ns
 - Frequency = 30.303 MHz

- Power Summary:
 - Total On Chip Power = 0.144 W
 - Junction Temperature = 26.7 °C
 - Thermal Margin = 58.3 °C (4.9 W)
 - Junction to Ambient Thermal Resistance = 11.5 °C/W

